

RXT2

Precalculus and Calculus

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A lab technician who works the night shift models a culture as $3 \cdot 2^x$ grams after x hours. The lab's freezer was replaced last week. When will the culture reach 48 grams?

- A. 8 hours
- B. 16 hours
- C. About 5.58 hours
- D. 4 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An economist whose report runs 40 pages uses $f(x) = (x^2 - 4) / (x^2 - x - 2)$ to model a cost ratio. Her manager wants the graph's features listed before the board meeting. Which description is correct?

- A. A vertical asymptote at $x = -1$ and a hole at $x = 2$
- B. Vertical asymptotes at $x = 2$ and $x = -1$
- C. A horizontal asymptote $y = 1$ and no vertical asymptote
- D. A vertical asymptote at $x = -2$ and a hole at $x = 2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which function has a hole at $x = 3$ and a vertical asymptote at $x = -2$?

- A. $f(x) = (x - 3) / (x + 2)$
- B. $f(x) = (x + 2) / [(x - 3)(x + 2)]$
- C. $f(x) = (x + 3) / [(x + 3)(x - 2)]$
- D. $f(x) = (x - 3) / [(x - 3)(x + 2)]$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retail analyst fits $f(x) = -2x^5 + 40x^4 + 3$ to sales data from a survey of 60 stores. What happens to $f(x)$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$?

- A. $f(x) \rightarrow -\infty$ in both directions
- B. $f(x) \rightarrow \infty$ as $x \rightarrow \infty$, and $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$
- C. $f(x) \rightarrow -\infty$ as $x \rightarrow \infty$, and $f(x) \rightarrow \infty$ as $x \rightarrow -\infty$
- D. $f(x) \rightarrow \infty$ as $x \rightarrow \infty$, because 40 is the largest coefficient

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A tide model for a small harbor with 80 boat slips is $y = 3 \sin(4x) + 1$, where x is in hours. The harbor master updates it every spring. What is the period of the model?

- A. 2π hours
- B. 8π hours
- C. 3 hours
- D. $\pi/2$ hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A sinusoid $y = \sin(Bx)$ with $B > 0$ has a period of 6. What is B ?

- A. 12π
- B. 6
- C. $\pi/3$
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A patient receives 80 mg of a drug with a half-life of 6 hours. The pharmacy that supplies the clinic is open 24 hours. How much of the drug remains after 18 hours?

- A. About 26.7 mg
- B. 10 mg
- C. 5 mg
- D. 40 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A population doubles every 5 years. How long does it take to grow from 500 to 4,000?

- A. 8 years
- B. 15 years
- C. 35 years
- D. 40 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A shipping rate is $f(x) = kx + 1$ for $x < 3$ and $f(x) = x^2 - 2$ for $x \geq 3$, with x in pounds. The company ships to 12 states. What value of k makes f continuous at $x = 3$?

- A. $k = 2$
- B. Any k , because each piece is a polynomial.
- C. $k = 8/3$
- D. $k = 3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A 2 kg cart moves along a track with velocity $v(t) = t - 3$ meters per second. The track was built by students in a robotics club. What is the value of the integral of $v(t)$ from $t = 0$ to $t = 4$, the cart's displacement?

- A. 5 meters
- B. 4 meters
- C. -4 meters
- D. -12 meters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

The integral of f from 0 to 5 is 10, and the integral of f from 0 to 2 is 16. What is the integral of f from 2 to 5?

- A. -6
- B. 26
- C. 6, since an area is never negative
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An intern at a firm with 200 employees must compute the integral of $3x^2 + 2$ from $x = 1$ to $x = 3$ for a flow report due Monday. What is its value?

- A.** 29
- B.** 30
- C.** 36
- D.** 33

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A lab technician ~~who works the night shift~~ models a culture as $3 \cdot 2^x$ grams after x hours. ~~The lab's freezer was replaced last week.~~ When will the culture reach 48 grams?

Target word: solve for the exponent **Decides:** $3 \cdot 2^x = 48$ gives $2^x = 16 = 2^4$.

A	Inverted operation	Divides 48 by 3 and then by 2 instead of taking a logarithm.
B	Confusion set	16 is the value of 2^x , not the exponent.
C	Dropped element	Solves $2^x = 48$ and drops the coefficient 3.
D	Correct	Divide by 3: $2^x = 16$, so $x = 4$.

2. Answer A Concept 2

An economist ~~whose report runs 40 pages~~ uses $f(x) = (x^2 - 4) / (x^2 - x - 2)$ to model a cost ratio. ~~Her manager wants the graph's features listed before the board meeting.~~ Which description is correct?

Target word: holes vs. vertical asymptotes **Decides:** $f = (x - 2)(x + 2) / [(x - 2)(x + 1)]$; the $(x - 2)$ cancels.

A	Correct	The canceled factor gives a hole; the remaining $(x + 1)$ gives the asymptote.
B	Dropped element	Skips the cancel step, so the shared factor is mistaken for an asymptote.
C	Adjacent category	$y = 1$ is a horizontal asymptote, but $x + 1$ still gives a vertical one.
D	Confusion set	$x = -2$ is a zero of the numerator (an x-intercept), not an asymptote.

3. Answer D Concept 2 · transfer

Which function has a hole at $x = 3$ and a vertical asymptote at $x = -2$?

Target word: build the function from its features **Decides:** $(x - 3)$ must cancel; $(x + 2)$ must stay in the denominator.

A	Dropped element	Nothing cancels, so $x = 3$ is a zero, not a hole.
B	Confusion set	Swaps the roles: hole at -2 and asymptote at 3.
C	Inverted operation	Signs flipped: hole at -3 and asymptote at 2.
D	Correct	$(x - 3)$ cancels to a hole at 3; $(x + 2)$ leaves an asymptote at -2 .

4. Answer C Concept 3

A retail analyst fits $f(x) = -2x^5 + 40x^4 + 3$ to sales data ~~from a survey of 60 stores~~. What happens to $f(x)$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$?

Target word: end behavior **Decides:** Leading term $-2x^5$: odd degree, negative coefficient.

A	Confusion set	That is even-degree behavior.
B	Inverted operation	That is the positive-leading-coefficient pattern.
C	Correct	An odd-degree term with a negative coefficient falls right and rises left.
D	Wrong stem number	The 40 belongs to a lower-degree term; for large $ x $ the x^5 term dominates.

5. Answer D Concept 4

A tide model for a small harbor with 80 boat slips is $y = 3 \sin(4x) + 1$, where x is in hours. The harbor master updates it every spring. What is the period of the model?

Target word: period **Decides:** Period = $2\pi / B$ with $B = 4$.

A	Dropped element	Drops the effect of the 4 inside the sine.
B	Inverted operation	Multiplies 2π by 4 instead of dividing.
C	Confusion set	3 is the amplitude, not the period.
D	Correct	$2\pi / 4 = \pi/2$.

6. Answer C Concept 4 • transfer

A sinusoid $y = \sin(Bx)$ with $B > 0$ has a period of 6. What is B ?

Target word: solve for B from the period **Decides:** $6 = 2\pi / B$, so $B = 2\pi / 6$.

A	Inverted operation	Multiplies 2π by 6.
B	Wrong stem number	That is the period, not B .
C	Correct	$B = 2\pi / 6 = \pi/3$.
D	Dropped element	Divides 6 by 2 and drops π .

7. Answer B Concept 5

A patient receives 80 mg of a drug with a half-life of 6 hours. The pharmacy that supplies the clinic is open 24 hours. How much of the drug remains after 18 hours?

Target word: amount remaining **Decides:** $18 / 6 = 3$ half-lives: $80 \times (1/2)^3$.

A	Adjacent category	Divides 80 by 3, treating decay as linear.
B	Correct	$80 \rightarrow 40 \rightarrow 20 \rightarrow 10$.
C	Camouflage	Uses the pharmacy's 24 hours: four half-lives.
D	Dropped element	Applies only one half-life.

8. Answer B Concept 5 • transfer

A population doubles every 5 years. How long does it take to grow from 500 to 4,000?

Target word: solve for time **Decides:** $4,000 / 500 = 8 = 2^3$, so 3 doubling times.

A	Wrong stem number	8 is the growth factor, not a time.
B	Correct	$3 \text{ doublings} \times 5 \text{ years} = 15$.
C	Adjacent category	Linear thinking: adds 500 every 5 years.
D	Inverted operation	Multiplies the growth factor 8 by 5 instead of using the number of doublings.

9. Answer A Concept 6

A shipping rate is $f(x) = kx + 1$ for $x < 3$ and $f(x) = x^2 - 2$ for $x \geq 3$, with x in pounds. ~~The company ships to 12 states.~~ What value of k makes f continuous at $x = 3$?

Target word: solve for k **Decides:** The pieces must meet: $3k + 1 = 3^2 - 2 = 7$.

A	Correct	$3k + 1 = 7$ gives $k = 2$.
B	Absolute language	Continuous pieces can still jump where they meet.
C	Inverted operation	Adds 1 instead of subtracting it: $(7 + 1) / 3$.
D	Wrong stem number	3 is the break point, not the slope.

10. Answer C Concept 7

~~A 2-kg cart~~ moves along a track with velocity $v(t) = t - 3$ meters per second. ~~The track was built by students in a robotics club.~~ What is the value of the integral of $v(t)$ from $t = 0$ to $t = 4$, the cart's displacement?

Target word: net displacement **Decides:** Antiderivative $t^2/2 - 3t$ evaluated from 0 to 4 is $8 - 12$.

A	Adjacent category	That is total distance $(4.5 + 0.5)$, not net displacement.
B	Inverted operation	Right size, wrong sign.
C	Correct	$8 - 12 = -4$; the 4.5 below the axis outweighs the 0.5 above.
D	Dropped element	Drops the $t^2/2$ term.

11. Answer A Concept 7 · transfer

The integral of f from 0 to 5 is 10, and the integral of f from 0 to 2 is 16. What is the integral of f from 2 to 5?

Target word: solve for a piece of the interval **Decides:** Integral from 0 to 5 = integral from 0 to 2 + integral from 2 to 5.

A	Correct	$10 - 16 = -6$.
B	Inverted operation	Adds the pieces instead of subtracting.
C	Absolute language	Integrals give signed area; -6 means f is mostly below the axis there.
D	Wrong stem number	That is the whole interval.

12. Answer B Concept 8

~~An intern at a firm with 200 employees~~ must compute the integral of $3x^2 + 2$ from $x = 1$ to $x = 3$ for a flow report ~~due Monday~~. What is its value?

Target word: definite integral **Decides:** $F(x) = x^3 + 2x$; $F(3) - F(1)$.

A	Confusion set	That is $f(3) = 27 + 2$, the integrand's value.
B	Correct	$F(3) = 33$ and $F(1) = 3$; $33 - 3 = 30$.
C	Inverted operation	Adds $F(1)$ instead of subtracting it.
D	Dropped element	Stops at $F(3)$ and drops $F(1)$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Solving exponential equations	Isolate the exponential first (divide out the coefficient), then take a logarithm or match bases.	Taking the log before dividing out the coefficient, or stopping at the value of the power.	A number times b^x equals a constant.
2	Holes vs. vertical asymptotes	Factor and cancel: a canceled factor gives a hole; a factor left in the denominator gives a vertical asymptote.	Calling every zero of the denominator an asymptote.	A rational function whose numerator and denominator share a factor.
3	Polynomial end behavior	The leading term alone decides end behavior: its degree (odd or even) and the sign of its coefficient.	Letting a larger coefficient on a lower-degree term decide.	"As $x \rightarrow \infty$ " or "as $x \rightarrow -\infty$ " with a polynomial.
4	Period and amplitude of sinusoids	For $y = A \sin(Bx) + D$, amplitude = $ A $ and period = $2\pi / B $.	Multiplying 2π by B , or reading B as the period.	A coefficient inside the sine or cosine.
5	Exponential growth and decay models	Amount = initial $\times (1/2)^{(t / \text{half-life})}$ or initial $\times 2^{(t / \text{doubling time})}$; change is by a constant factor, not a constant amount.	Subtracting or adding a fixed amount each period (linear thinking).	A half-life or doubling time and an elapsed time.
6	Continuity at a point	f is continuous at a when $f(a)$ exists, the limit exists, and they are equal; for piecewise functions the pieces must meet.	Assuming a function made of continuous pieces is continuous everywhere.	"Find k so that f is continuous."
7	Definite integral as net signed area	Area below the x -axis counts as negative; the integral gives net change, not total area.	Adding all areas as positive.	An integrand that changes sign on the interval.
8	Fundamental Theorem of Calculus	The integral from a to b of f equals $F(b) - F(a)$ for any antiderivative F .	Stopping at $F(b)$, adding $F(a)$, or evaluating f instead of F .	A definite integral of a polynomial with numeric bounds.

Course scope (public description): Precalculus and Calculus provides instruction in precalculus and calculus and applies them to examples found in both mathematics and science. Topics in precalculus include principles of trigonometry, mathematical modeling, and logarithmic, exponential, polynomial, and rational functions. Topics in calculus include conceptual knowledge of limit, continuity, differentiability, and integration.